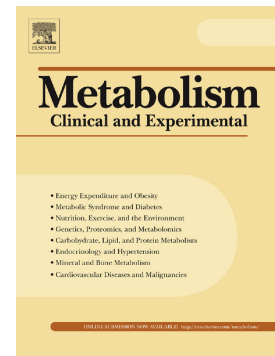


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The mitochondrial TSPO ligand Atriol mitigates metabolic-associated steatohepatitis by downregulating CXCL1

Yuchang Li¹, Liting Chen¹, Chantal Sottas¹, Mahima Chandrakant Raul¹, Nrupa Dinesh Patel¹, Janaki Ramulu Bijja¹, S. Kaleem Ahmed¹, Audrey Kapelanski-Lamoureux^{2,3}, Anthoula Lazaris³, Peter Metrakos^{2,3,4}, Alexander Zambidis¹, Shefali Chopra⁵, Meng Li⁶, Go Sugahara^{7,8}, Takeshi Saito^{7,9}, and Vassilios Papadopoulos^{1*}

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Abstract

Background and Aims:

The mitochondrial translocator protein (TSPO, 18kDa) is pivotal in binding cholesterol and facilitating its transfer from the outer to the inner mitochondrial membrane. Atriol is a TSPO ligand disrupting cholesterol binding by targeting the cholesterol-recognition amino acid consensus domain. Prior research has shown that TSPO deficiency improved metabolic-associated steatohepatitis (MASH). We hypothesized that Atriol may have the potential to alleviate MASH.

Methods and Results

In vitro cell culture studies revealed that Atriol treatment effectively mitigated MASH by restoring mitochondrial function, inhibiting the NF- κ B signaling pathway, and reducing hepatic stellate cell (HSC) activation. SD male rats were fed a GAN diet for 10 months to induce MASH. During the final two weeks of feeding, rats received intraperitoneal Atriol administration daily. Atriol treatment significantly ameliorated MASH by reducing lipid accumulation, diminishing hepatic lobular inflammation and fibrosis, decreasing cell death, and inhibiting excessive bile acid synthesis. Moreover, Atriol restored mitochondrial function in primary hepatocytes isolated from MASH rats. In search of the mechanism(s) governing these effects, we found that Atriol downregulated the proinflammatory chemokine CXCL1 through the NF- κ B signaling pathway or via myeloperoxidase (MPO) in HSCs and Kupffer cells. Additionally, in vitro, studies further suggested that CXCL1 treatment induced dysfunctional mitochondria, inflammation, HSCs activation, and macrophage migration, whereas Atriol countered these effects. Finally, the mitigating effects of Atriol on MASH were reproduced by pharmacological inhibition of NF- κ B or MPO and neutralization of CXCL1.

Conclusion:

Atriol ameliorates MASH both in vitro and in vivo, demonstrating its potential therapeutic benefits in managing MASH.

Keywords:

ADME (absorption, distribution, metabolism, and excretion), pharmacokinetic profile, crown-like structures, fibrosis, myeloperoxidase, proinflammation, mitochondrial function

Abbreviations:

ACOX1(acyl-coenzyme A oxidase 1), ADME (absorption, distribution, metabolism, and excretion), ACTA2 (alpha-smooth muscle actin), ALT (alanine aminotransferase), AST (aspartate aminotransferase), Atriol (3,17,19-androsten-5-triol), CCR (CC Motif Chemokine Receptor), CEL (Celastrol), CM (culture media), Col1a1 (collagen type I α 1), Col1a2 (collagen type I α 2), CPT1A (carnitine palmitoyltransferase 1A), CXCR (CXC Motif Chemokine Receptor), CXCL1 (CXC motif chemokine ligand 1), CYP27A1(cytochrome P450 family 27 subfamily A member 1), DGAT(diacylglycerol O-acyltransferase 1), ELOV6 (elongation of very long chain fatty acids protein 6), FATP(fatty acid transport protein), FGF21 (Fibroblast growth factor 21), GAN (Gubra-Amylin MASH diet), HFD (high-fat diet), IL1 β (Interleukin 1 β), KC (Kupffer cells), LPS (lipopolysaccharide), MASLD (metabolic dysfunction-associated steatotic liver disease), MCDD (methionine-choline-deficient diet), MPO (myeloperoxidase), MMP-2 (matrix metalloproteinase 2), NEFAs (Non-esterified fatty acids), NLRP3 (NLR family pyrin domain containing 3), NF- κ B (nuclear factor kappa B), OA (and oleic acid), PAI-1 (plasminogen activator inhibitor 1), PK (pharmacokinetic), PMA (phorbol 12-myristate 13-acetate), ROS (reactive oxygen species), SCD (stearoyl-Coenzyme A desaturase), SREBF (sterol regulatory element-binding transcription factor), TIMP1(Tissue inhibitor of metalloproteinases-1), TLR4 (toll-like receptor 4), TSPO (translocator protein ,18 kDa).

1. Introduction

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a common condition characterized by hepatic fat accumulation without significant alcohol intake¹. MASLD encompasses simple steatosis (SS) and metabolic-associated steatohepatitis (MASH), which involves inflammation, fibrosis, and dysfunctional mitochondria¹. In severe cases, MASH can progress to liver cirrhosis and hepatocellular carcinoma

(HCC)². While weight reduction strategies show positive outcomes in improving MASLD, there are limited options for the treatment of MASH³. Drug development efforts primarily target lipogenic synthesis, inflammation, fibrosis, insulin sensitivity, and oxidative stress^{4,5}. Promising candidates like obeticholic acid (OCA, Intercept), a farnesoid X receptor (FXR) agonist; resmetirom (Madrigal Pharmaceuticals), a liver-targeted thyroid hormone receptor (TR) β agonist; and Pegzofermin, a long-acting glycopegylated FGF21 analog, all showing efficacy in mitigating MASH but with undesirable side effects^{3,6,7}. Understanding the metabolic mechanisms of MASH is crucial, and pharmacological interventions targeting multiple pathways may offer superior efficacy in managing MASH.

Activation of the transcription factor nuclear factor kappa B (NF- κ B) is increased in patients with MASH⁸. Once activated, NF- κ B translocates to the nucleus and promotes transcription of genes that mediate inflammation, apoptosis, and fibrosis in MASH⁹. Overexpression or ligand-activated PPAR α or FXR has been shown to suppress NF- κ B signaling-regulated inflammation^{10,11}. The CXC motif chemokine ligand 1 (CXCL1), a potent chemokine that attracts neutrophils to inflamed or infected sites, is regulated by the NF- κ B signaling pathway^{12,13}. Forced expression of CXCL1 in rat liver led to neutrophil infiltration, causing hepatic damage indicated by elevated ALT and AST levels¹⁴. Additionally, CXCL1 overexpression in the liver alone promotes steatosis-to-MASH progression in high-fat diet (HFD)-fed mice by inducing neutrophil infiltration, oxidative stress, and stress kinases¹⁵. The elevated CXCL1 in HFD-induced obese mice correlated with myeloperoxidase (MPO)-positive neutrophils¹⁶. MPO, a potent inflammatory enzyme, is increased in patients with MASLD¹⁷. MPO-deficient mice displayed reduced liver inflammation, fibrosis, and hepatocyte injury in response to HFD¹⁸. MPO promotes oxidative stress and tissue damage in the liver through reactive oxygen species (ROS) production, contributing to MASH progression¹⁹. Thus, targeting MPO, CXCL1, or associated signaling pathways may offer a valuable therapeutic approach for MASH (ClinicalTrials.gov).

TSPO, a high-affinity cholesterol-binding outer mitochondrial membrane protein, plays a crucial role in regulating cellular energy metabolism and

responding to cellular stress^{20,21}. Recent studies indicate that TSPO expression is increased in liver tissue of MASH patients and animal models²²⁻²⁴. TSPO overexpression is known to be associated with inflammation^{25,26}, playing a role in the transition from SS to MASH. Conversely, loss of TSPO improved methionine-choline-deficient diet (MCDD)-induced MASH²³, suggesting TSPO is a potential therapeutic target for suppressing MASH. In this study, we explored the impact of 3,17,19-androsten-5-triol (Atriol), a TSPO ligand binding to the cholesterol recognition/interaction amino acid consensus (CRAC) motif^{23,27}, on advanced MASH stages in Gubra-Amylin MASH diet (GAN)-induced rats²⁸. The investigation suggests that Atriol is a promising therapeutic candidate with multiple benefits, effectively reducing liver steatosis, fibrosis, inflammation, and cell death and restoring mitochondrial function in MASH. These positive effects are achieved by regulating NF- κ B, MPO, and CXCL1 expression in various liver cell types.

2. Materials and methods

2.1 Experimental methods

Detailed materials and methods are provided as **Supplemental Materials and Methods**.

2.2 Statistical analysis

Data analyses were performed with GraphPad Prism 10.2.0 software. Analysis was performed using a student's t-test between two comparisons or one-way ANOVA with Turkey's correction for multiple comparisons. $p < 0.05$ was considered statistically significant.

3. Results

3.1 TSPO-ligand Atriol reverses the MASH phenotype in vitro

After observing the beneficial effects of TSPO deletion in MCDD-induced MASH²³, we investigated the impact of inhibiting TSPO by 3,17,19-androsten-5-triol (Atriol) (Fig.1A), a ligand specifically targeting the CRAC motif in TSPO²⁷, on MASH in vitro. To initiate this investigation, we evaluated the in vitro ADME (absorption, distribution, metabolism,

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and excretion) properties and in vivo pharmacokinetic (PK) analysis in rats of Atriol. Most of the ADME and physicochemical characteristics met the target criteria. The rat liver microsome stability (RLM $t_{1/2}$ min) and liver microsome clearance (RLM-in vitro CL_{int} $\mu\text{L}/\text{min}/\text{mg}$ protein) were within moderate ranges. Aqueous solubility, LogD and plasma protein binding (% of PPB) were all within the target range (Table 1a). The in vivo PK analysis showed i.p-20 mg/kg dose ($AUC = 2369 \text{ h}\cdot\text{ng}/\text{mL}$; $t_{1/2} = 1.39 \text{ h}$; %F = 70.8) is the optimal dose for administration (Table 1b). Taken together these data suggested that Atriol is a viable drug candidate for further assessment.

First, we evaluated Atriol's interaction with TSPO in Huh-7 cells using Cellular Thermal Shift Assay (CETSA), a technique for assessing the binding affinity and interaction between a drug and its target protein in live cells^{29,30}. The results showed substantial TSPO stabilization with Atriol, evidenced by a thermal shift of up to 10°C (Fig 1B, Fig. S1A, B), indicating a direct interaction between TSPO and Atriol.

We then induced MASH with lipopolysaccharide (LPS) and oleic acid (OA) in Huh-7 or THP-1 derived macrophages, resulting in NF- κB p65 nuclear translocation. Exposing cells to different Atriol concentrations (10 μM , 50 μM , 100 μM) progressively reduced nuclear p65 levels either in Huh-7 or THP-1 cells (Fig. 1C), indicating Atriol's potential to significantly diminish inflammation during MASH. Subsequently, we selected 100 μM Atriol for later in vitro experiments.

Next, we assessed mitochondrial function in humanized liver chimeric mouse-derived human hepatocytes (HLCM-HH) treated with LPS+OA, with or without Atriol. Seahorse analysis revealed that Atriol treatment restored the reduced oxygen consumption rate (OCR) caused by LPS+OA, resulting in significantly higher OCR of mitochondrial basal respiration, maximal respiration, and ATP production (Fig. 1D). Similar restoration was observed in Huh-7 cells treated with LPS+OA with or without Atriol (Fig. S1C), indicating that Atriol restores mitochondrial function in simulated MASH cells.

Finally, we activated HSCs in LX-2 cells with TGF β , upregulating fibrotic markers ACTA2 and COL1A1. However, this upregulation was inhibited

by Atriol treatment (Fig. 1E, 1F). Additionally, in rat BSC cells (a rat HSC cell line) exposed to TGF β , Atriol reduced ACTA2 upregulation (Fig. S1D, 1E). Obeticholic acid (OCA), a potent FXR agonist, served as a control treatment³¹. These data indicate that Atriol alleviates HSCs activation.

3.2 Atriol ameliorates the damaged structure of the liver in MASH

Male WT SD rats were fed a GAN diet for 43 weeks, inducing the MASH phenotype, with a low-fat diet as a control. In the final two weeks, rats received daily intraperitoneal administration of Atriol (20 mg/kg) or DMSO, resulting in four groups: LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol (Fig. 2A). After 43 weeks, no significant differences in food intake were observed between LFD and GAN diets or during Veh or Atriol administration. Rats on the GAN diet displayed an enlarged liver size and an increased liver weight-to-body weight ratio (LW/BW) (Anatomy in Fig. 2B, 2C), indicative of fatty livers. Interestingly, under an LFD diet, Atriol led to an elevation in plasma TAG and a potential increase in insulin, indicating TAG accumulation. This aligns with our earlier discovery that Atriol temporarily exacerbates steatosis in hepatocytes²³. Despite this, Atriol significantly lowered LW/BW ratios and plasma concentrations of ALT, AST, cholesterol, TAG, insulin, NEFA, and liver TAG (Fig. 2C), which were elevated with GAN diet plus vehicle.

Following GAN diet, the livers exhibited typical MASH features, including steatosis (H&E and Oil Red O (ORO) staining in Fig. 2B, black arrow in Fig. S2A), hepatocyte ballooning (red arrow, Fig. S2A), inflammatory cell infiltration (white arrow, Fig. S2A), Mallory-Denk bodies (MDBs, green arrow, Fig. S2B), and atypical nuclei with condensed nucleoli (yellow arrow, Fig. S2C). These features were either diminished or reduced after Atriol treatment (Fig. 2B, Fig. S2D). Furthermore, the bridging fibrosis caused by GAN diet was also reduced following Atriol treatment (Fig. 2B, Fig. S2D). These observations were confirmed through a blinded assessment using the NAFLD/MASLD activity score system by a pathologist (Table S1, Fig. S2E). These findings suggest that Atriol could potentially reverse damaged liver structure in MASH.

Considering that the primary source of TAG is NEFAs from white adipose tissue (WAT), which contribute to lipid accumulation in hepatocytes³², we investigated the impact of Atriol on WAT in MASH. Morphological analysis revealed GAN-induced adipocyte hypertrophy was alleviated by

Atriol treatment (white arrows in H&E staining, Fig. 2D). Additionally, the pronounced chronic inflammation marked by crown-like structures (CLS) (red arrowheads in H&E staining, Fig. 2D), which stained F4/80 positive in GAN-fed rats (white arrow in Fig. 2D) (Fig. S2F), was reduced with Atriol treatment. The upregulated levels of inflammation markers CD11B and PAI-1 following the GAN diet were attenuated upon Atriol treatment. Conversely, the anti-inflammatory markers Adiponectin and IL-10 were found to be elevated in the GAN+Atriol compared to the GAN+Veh group (Fig. S2G). This observation implies that the inflammation in WAT was intensified in MASH but was alleviated by Atriol treatment.

3.3 Atriol reduces liver inflammation, fibrosis, cell death, de novo lipogenesis, and fatty acid transporter markers, while enhances mitochondrial function indicators in MASH

Next, we investigated the key markers responsible for the cellular characteristics during the transition from SS to MASH. The markers for inflammation (CD11B), fibrosis (ACTA2, mature TGF- β 1), and apoptosis (cleaved caspase3, C-CASP3) in livers from the GAN+Atriol group were significantly reduced compared to GAN+Veh (Fig. 3A). This was supported by decreased levels of inflammation markers (F4/80, CD11B, NF- κ B) and fibrosis marker ACTA2 after Atriol treatment in MASH (Fig. 3B). TUNEL immunofluorescent staining revealed cell death induced by the GAN diet was significantly reduced by Atriol treatment (Fig 3C). Additionally, qPCR analysis indicated that GAN diet mRNA upregulation of markers for fibrosis (*Col1a1*, *Col1a2*, *Acta2*, *Tgf β* , *Timp1*), inflammatory signaling (*Tnfa*, *Il1 β* , *Nlrp3*, *Cxcl1*, *F4/80*), de novo lipogenesis (DNL) (*Acc1*, *Elov6*, *Fasn*, *Dgta1*, *Scd1*), and fatty acid transporter *Fatp5* and *Cd36*, along with the downregulation of *Ppara*, a mitochondrial β -oxidation marker, and *Pgc1 α* , a mitochondrial biogenesis marker, were all successfully reversed with Atriol treatment (Fig. 3D). Collectively, our observations demonstrated that Atriol treatment effectively halted the advancement of MASH by modulating liver inflammation, fibrosis, cell death, DNL, and lipid transport, while promoting mitochondrial biogenesis and fatty acid oxidation (FAO).

3.4 Atriol maintains bile acid homeostasis and restores mitochondrial function without increasing ROS production in MASH in vivo

We previously demonstrated that TSPO deficiency mitigated MCDD-induced MASH by reduced BA production via upregulated FXR in the

livers²³. Here, we investigated whether the mitigation of Atriol on GAN-induced MASH through BA synthesis. Firstly, we observed an increase in TSPO and CYP27A1 levels with GAN but were significantly reduced by Atriol treatment (Fig. S3), suggesting Atriol inhibited the increased TSPO level in MASH and was involved in the BA production. Meanwhile, FXR levels were reduced with GAN but approximately 2 times higher after Atriol treatment (Fig. 4A), indicative of the role of Atriol regulation of FXR in MASH. We then proceeded to investigate if changes occurred in the composition of serum BAs. The results showed increased levels of glycocholic acid (GCA), glycochenodeoxycholic acid (GCDCA), glycodeoxycholic acid (GDCA), and taurodeoxycholic acid (TDCA) in GAN+Veh, which were notably decreased following Atriol treatment (Fig. 4B). Further, we assessed the impact of these altered BAs on fibrogenesis by treatment of LX-2 cells with each BA individually (GCA, GCDCA, GDCA, or TDCA) or in combination (BAX: GCA+GCDCA+GDCA+TDCA). qPCR results revealed a significant upregulation in fibrotic markers such as *TGFβ*, *COL1A1*, and *ACTA2* with either single bile acid or combinatory BAX treatments (Fig. S4A-4E). Moreover, these individual BA treatments significantly impaired mitochondrial function in Huh7 cells by reducing basal, maximal respiration, and ATP production, while Atriol restored them back to normal levels (Fig. 4C). Together, these findings suggest that altered BA synthesis may link to liver fibrogenesis and dysfunctional mitochondria, while Atriol can maintain BA homeostasis to mitigate MASH.

Our in vitro data suggested Atriol may restore mitochondrial function in MASH cells (Fig. 1A, 1B). We further assessed the role of Atriol on mitochondrial function in MASH in vivo. PPARα and its downstream CPT1A, both pivotal elements for FAO^{33, 34}, decreased in livers with GAN diet but exhibited higher expression after Atriol treatment in MASH (Fig. 4A, Fig. S4F). Further, we isolated hepatocytes from GAN+Veh and GAN+Atriol groups to evaluate mitochondrial functions. The results showed significantly higher basal and maximal respiration and ATP production in GAN+Atriol compared to GAN+Veh (Fig. 4D). Consistently, higher expression of CV-ATP5A and CIII-UQCRC2, components of the oxidative phosphorylation (OXPHOS) complex, was observed in primary hepatocytes in GAN+Atriol than GAN+Veh (Fig. 4E). Additionally, PGC1α

and PPAR α levels were significantly higher, whereas pro-TGF β was nearly undetectable in the GAN+Atriol group compared to GAN+Veh (Fig. 4E). These data collectively indicate that Atriol effectively restored impaired mitochondrial function in MASH in vivo.

To investigate whether potentially increased ROS production arising from elevated FAO could interfere with the mitochondrial electron transport chain (ETC)³⁵, we analyzed the expression of ETC complexes in liver tissue. The results indicated 1.5~2-fold higher expression of CV-ATP5A and CIII-UQCRC2 in GAN+Atriol compared to GAN+Veh (Fig. 4A, Fig. S4F), suggesting no ETC impairment by Atriol in MASH. Moreover, we evaluated hepatic ROS by immunofluorescence staining for 4-hydroxynonenal (4-HNE), an end-product of peroxidation of membrane N-6-polyunsaturated fatty acids, and malonaldehyde (MDA), the aldehyde products of lipid peroxidation (LPO), both used as sensitive markers of oxidative damage and LPO itself^{36,37}. The results showed that the signals of 4-HNE (white arrow) or MDA (white arrowhead) were substantially increased after GAN diet but were eliminated with Atriol treatment.

3.5 CXCL1 is downregulated by Atriol to ameliorate MASH

To understand how Atriol improves MASH, we analyzed plasma cytokine profiles. The results showed that the levels of proinflammatory chemokine (C-X-C motif) ligand 1 (CXCL1), intercellular adhesion molecule 1 (ICAM1), CC motif chemokine ligand 3 (CCL3), tissue inhibitor of metalloproteinase-1 (TIMP-1), and vascular endothelial growth factor (VEGF) were reduced by approximately 2.5, 1.2, 2.2, 1.6, and 1.2 fold, respectively, in GAN+Atriol compared to GAN+Veh. At the same time, no such changes were observed between LFD+Veh and LFD+Atriol groups (Fig. 5A). As reported, CXCL1 is prominently upregulated in the liver of MASH patients and in mice exposed to an HFD or a choline-deficient amino acid-defined (CDAA) diet³⁸⁻⁴⁰. Thus, we explored the potential interaction between Atriol and CXCL1 in MASH. Firstly, we assessed the protein levels of CXCL1 and myeloperoxidase (MPO), a peroxidase enzyme responsible for inflammation and oxidative stress, which was shown to be upregulated in MASH patients¹⁷. Immunoblot and immunofluorescence staining indicated both CXCL1

and MPO were upregulated with a GAN diet but were reduced after Atriol treatment (Fig. 5B,5C).

Secondly, considering the crucial roles of MPO and CXCL1 in the pathogenesis of MASH, we investigated whether MPO can regulate CXCL1. We preincubated Huh-7, LX-2, or THP-1 derived macrophages with recombinant human MPO protein, along with 4-aminobenzoic acid hydrazide (ABAH, an irreversible MPO inhibitor)⁴¹ or taurine (an hypochlorous acid [HOCl] scavenger)⁴². qPCR showed no significant change in *CXCL1* expression amongst Veh, MPO, MPO+ABAH, or MPO+Taurine treatments in Huh-7 cells (Fig. 5D, left). However, in LX-2 cells *CXCL1* level was significantly increased after preincubation with MPO but decreased after ABAH or taurine treatment (Fig. 5D, middle), confirmed by immunofluorescence staining (green signal in Fig. 5E). In THP-1 cells, *CXCL1* level did not change after MPO treatment but significantly decreased after ABAH or Taurine treatment (Fig. 5D, right, and green signal in Fig. 5E). Interestingly, COL1A1 (red signal in Fig. 5E) was found to be upregulated by MPO but inhibited by ABAH or taurine treatment, suggesting MPO regulation of CXCL1 in HSCs. In THP-1 cells, the F4/80 signal mirrored that of CXCL1, showing no alteration post-MPO treatment, yet exhibiting a significant reduction following ABAH or Taurine treatment (Fig. 5E). Further, we treated LX2 with TGF β with or without Atriol and further qPCR analysis showed that Atriol significantly reduced the upregulated MPO expression by TGF β (Fig. 5F). Moreover, we examined CXCL1 levels in conditioned medium (CM) from LX-2 or THP-1 cells treated with MPO protein, with or without ABAH. ELISA analysis revealed that CXCL1 levels in MPO-treated CM were significantly elevated compared to Veh but decreased after ABAH treatment in LX2 CM (Fig. 5G). In the CM from THP-1 cells, CXCL1 level did not change after MPO treatment but significantly reduced by ABAH (Fig. 5G). These data suggest MPO selectively regulates CXCL1 in HSCs and Kupffer (KC) but not in hepatocytes.

To further investigate the role of MPO in vivo, we administered ABAH or vehicle to GAN-induced MASH rats for 4 weeks. GAN+ABAH rats exhibited reduced liver inflammation, fibrosis, and lipid accumulation compared to GAN+Veh controls (Fig. S5A-5C). Serum MPO levels were significantly decreased in GAN+ABAH compared to GAN+Veh (Fig.

S5D). However, serum CXCL1 levels showed minimal variation (Fig. S5E), suggesting MPO's selective influence on CXCL1. Immunofluorescent staining in the livers from GAN+ABAH and GAN+Veh groups revealed CXCL1 expression in HSCs and KCs, co-localizing with HSC marker Desmin, and KC marker F4/80 (green signal in Fig. 5H). GAN+ABAH exhibited reduced CXCL1 expression in HSCs or in KCs compared to GAN+Veh (Fig. 5H). ACTA2 was also significantly decreased in GAN+ABAH livers (Fig. 5I). Additionally, GAN+ABAH showed lower liver inflammation, evidenced by reduced CD11B and F4/80 levels compared to GAN+Veh (Fig. S5F, S5G). Moreover, there was a significant reduction in F4/80-positive CLS in WAT in GAN+ABAH compared to GAN+Veh (Fig. S5H). Overall, these findings strongly support that Atriol not only downregulated CXCL1 in MASH but also simultaneously reduced MPO levels, leading to a further decrease in CXCL1 within HSCs or KCs. Consequently, this dual effect contributes to the alleviation of MASH.

3.6 CXCL1 impairs mitochondrial function, activates HSCs, induces inflammation, and promotes macrophage infiltration, whereas Atriol counters these effects

Through analysis of the scRNA-seq GSE136103 dataset from healthy and fibrotic patients, CXCL1 was found to be expressed in the three primary cell types of the liver- hepatocytes, HSCs, and KCs⁴³ (Fig. S6A). Next, we examined the colocalization of CXCL1 in three liver cell types in normal, steatosis, and MASH patient livers, validated by H&E staining (Fig. 6A) and independent evaluation by a pathologist using the MASLD activity score system (Table S2). Immunofluorescence staining confirmed CXCL1 colocalization with the hepatocyte marker asialoglycoprotein receptor 1 (ASGR1), the KC marker CD68, and the HSC marker DESMIN (Table S3). Notably, in MASH compared to normal and SS, the CXCL1 signal was increased by about 50% and 16% (Fig. 6A). Secondly, we treated freshly isolated HLCM-HH with CXCL1, with or without Atriol. Seahorse assay results demonstrated that basal respiration, maximal respiration, and ATP production were reduced by CXCL1 treatment while being restored with the addition of Atriol (Fig. 6B), suggesting Atriol can counteract the adverse effects on mitochondrial function by CXCL1. Thirdly, we subjected human LX-2 cells or rat BSC to CXCL1 treatment, with or without Atriol. CXCL1 significantly increased

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the expression of *ACTA2* and *COL1A1*, whereas the addition of Atriol reduced their expression (Fig. 6C), which was confirmed by *COL1A1* expression using immunofluorescence staining (Fig. 6D). This supports the conclusion that Atriol effectively mitigates the activation of HSCs induced by CXCL1. Subsequently, we preincubated THP-1 monocyte cells with phorbol 12-myristate 13-acetate (PMA) to induce differentiation to macrophages, followed by treatment with CXCL1, with or without Atriol. Immunoblot and immunofluorescent staining analyses revealed a significant increase in TNF α levels with CXCL1 treatment, which was reduced after Atriol treatment (Fig. 6E-6F), suggesting Atriol may inhibit the proinflammatory response induced by CXCL1 in THP-1 macrophages.

Next, we evaluated the mutual interaction between hepatocytes, HSC, and THP-1 macrophages utilizing a co-culture system. To assess the influence of CXCL1 with or without Atriol, Huh-7 cells were treated with CXCL1, with or without Atriol in the upper chamber, while the lower chamber contained LX-2 cells. qPCR analysis of LX-2 cells showed significant upregulation of *ACTA2*, *COL1A1*, *TIMP-1*, *TGF β 1*, and *MMP-2* after CXCL1 treatment, which was reversed upon Atriol treatment (Fig. S6B), suggesting a paracrine effect from Atriol-treated hepatocytes, mitigating the CXCL1-induced activation of HSCs. We reversed the arrangement, so that the upper chamber contained LX-2 cells and was treated with CXCL1, with or without Atriol, while the lower chamber consisted of Huh-7 cells. qPCR analysis of Huh7 cells demonstrated significant upregulation of acetyl-CoA carboxylase expression (*ACC*), a key enzyme for DNL, which was reversed after Atriol treatment (Fig. S6C), indicating a paracrine effect of Atriol-treated HSCs, preventing CXCL1-induced fatty acid synthesis.

To explore the impact of CXCL1-treated Huh-7 hepatocytes on macrophage migration, THP-1 cells, preincubated with PMA for 3 days to induce differentiation, were placed in the upper chamber, while the lower chamber contained Huh-7 cells treated with CXCL1, with or without Atriol. Migrated macrophages to the insert membrane were fixed, stained, and photographed. The analysis revealed that macrophage migration was enhanced by CXCL1-treated Huh-7 cells, whereas Atriol treatment inhibited macrophage migration (Fig. 6G). Next, we performed

another co-culture experiment with the upper chamber containing Huh-7 treated with CXCL1, with or without Atritol, and the lower chamber containing THP-1-derived macrophages. qPCR analysis of the macrophages showed a substantial increase in inflammatory markers (*TNF α* , *IL1 β* , *CXCR2*, *CXCR3*, *CCR2*, and *CCR5*) with CXCL1 treatment. However, these increases were reversed by Atritol treatment, indicating that Atritol prevented macrophage migration induced by CXCL1 (Fig. S6D). Taken together, these results suggest that CXCL1 treatment leads to impaired mitochondrial function, HSC activation, inflammation, and macrophage migration, while Atritol effectively prevents these effects.

3.7 Pharmacological inhibition of NF- κ B/CXCL1 axis or neutralizing CXCL1 effectively reproduces the mitigating effects of Atritol on MASH.

Given that Atritol substantially decreased CXCL1 levels in MASH, we examined whether CXCL1 plays a dominant role in Atritol's anti-MASH effects. Previous studies have demonstrated direct binding of NF- κ B to the CXCL1 promoter in both rat and human⁴⁴⁻⁴⁸. To confirm this, we conducted a luciferase assay in Huh7 cells by co-transfection of human NF- κ B p65 (RELA) and pGL3-CXCL1-Luc plasmids (Fig. S7A). In addition, we employed Celastrol (CEL), an NF- κ B inhibitor, to suppress CXCL1 activity induced by *TNF α* treatment in a dose-dependent manner (Fig. S7B). To examine the role of NF- κ B/CXCL1 axis in vivo, we daily administered CEL to GAN-fed rats for 3 weeks, which resulted in a significant 10% reduction in body weight (GAN+CEL vs. GAN+Veh, Fig. S7C). Histological analysis of liver tissue showed decreased inflammation, lipid accumulation, and fibrosis in GAN+CEL compared to GAN+Veh (Fig. S7D). In addition, a reduced number of inflammatory cells were observed in the adipocyte interstitium of WAT in GAN+CEL compared to the GAN+Veh animals (H&E staining in WAT in Fig. S7D). Immunoblot analysis of liver showed significant reductions in *TNF α* , CXCL1, and ACTA2 in GAN+CEL compared to GAN+Veh (Fig. 7A). Immunohistochemistry staining suggested reduced signals for nuclear NF- κ B (black arrow), F4/80, CD11B, and ACTA2 in GAN+CEL compared to controls (Fig. 7B). Further, we extracted nuclear/cytoplasm proteins from primary hepatocytes isolated from GAN+Veh and GAN+CEL. Immunoblot analysis revealed a notable reduction in the

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nuclear/cytoplasmic NF- κ B ratio upon CEL treatment (Fig. 7C). These data suggest that pharmacological inhibition of NF- κ B/CXCL1 axis can ameliorate MASH.

To further assess the role of CXCL1 in vivo, we intravenously administered a CXCL1-neutralizing antibody (CXCL1 neu) to rats with GAN-induced MASH (GAN+CXCL1 neu). Analysis indicated stable liver size, but GAN+CXCL1 neu rats showed substantial reductions in liver inflammation (red arrows), lipid accumulation, and fibrosis compared to GAN+Veh (Fig. S7E). Moreover, the inflammation markers (TNF α , IL1 β) and fibrosis marker (ACTA2) in liver tissue was decreased in GAN+CXCL1 neu but with induction of FXR and PPAR α compared to GAN+Veh (Fig. 7D), which was also confirmed by immunohistochemical staining of reduced F4/80, cytosol stabilization of NF- κ B P65, and reduced ACTA2 (Fig. 7E). Moreover, the levels of thiobarbituric acid-reactive substance (TBARS), an LPO marker, and dihydroethidium (DHE), an oxidative stress marker, significantly decreased after CXCL1 neu treatment compared to controls (Fig. 7F, Fig. S7F). qPCR analysis indicated reduced levels of fibrosis markers (*Col1a1*, *Acta2*, *Tgf β* , *Timp1*), inflammation markers (*Tnf α* , *Il1 β* , *F4/80*, *Cd68*, *Nlrp3*, *Tlr4*), chemokines and ligand markers (*Ccl2*, *Ccl20*, *Cxcl9*, *Cxcl12*, *Cd621/L-selectin*), and fatty acid transport/lipid synthesis markers (*Fatp1*, *Fatp5*, *Cd36*, *Srebf1*, *Srebf2*, *Fasn*, *Dgat1*, *Dgat2*, *Acc*, *Scd1*, *Elov6*) in GAN+CXCL1 neu rats compared to GAN-Veh (Fig. S7G). Conversely, mitochondrial biogenesis marker *Pgc1 α* and FAO markers (*Acox 1*, *Cpt1 α*) were significantly induced in the GAN+CXCL1 neu treated rats compared to GAN-Veh (Fig. S7G). Collectively, our findings suggest that reducing CXCL1 levels replicates the therapeutic effects seen with Atriol treatment, leading to an improvement in MASH.

4. Discussion

Atriol, a ligand targeting the cholesterol-binding domain of TSPO, improved MASH in rats by reducing hepatic lobular inflammation, fibrosis, cell death, and by restoring mitochondrial function. MASH pathogenesis is characterized by downregulated PPAR α and FXR,

resulting in increased NF- κ B signaling and CXCL1. CXCL1 treatment induces inflammation, fibrosis, and macrophage migration. Moreover, MPO is upregulated in MASH. Atriol counteracts these effects by targeting MPO and CXCL1. Pharmacologic inhibition of NF- κ B with CEL, MPO with ABAH, or CXCL1 neutralization has been shown to alleviate MASH effectively.

In vitro, Atriol disrupted cholesterol binding in hepatocytes and mirrored alterations observed in TSPO-deficient cells²³. Consistent with our current study, Atriol treatment induced TAG accumulation in the livers of rats on LFD. In this study, rats began experimental diets at 40 days, aging over the 10-month feeding period. Since lipid accumulation tends to increase with age⁴⁹, it was overtaken by accelerated accumulation in the GAN-fed group, leading to MASH. Notably, Atriol effectively prevented this lipid accumulation, as supported by lipid and biogenetic studies. Our findings suggest that Atriol plays a complex role in MASLD pathogenesis, initiating steatosis but rescuing MASH. Further investigation is needed to clarify its optimal administration timing for MASH prevention and treatment.

Notably, Atriol demonstrates the ability to modulate both MPO and CXCL1. Our in vitro and in vivo studies have mechanistically revealed Atriol's regulatory effect on CXCL1 via NF- κ B signal pathway in hepatocytes and macrophages. In contrast, MPO autonomously regulates CXCL1 expression in HSCs and Kupffer cells. Whether Atriol directly regulated CXCL1 in HSCs or Kupffer cells requires further investigation. It is important to note that our in vitro study indicated that MPO treatment did not elevate CXCL1 levels in PMA-induced macrophages derived from THP-1 cells. This observation aligns with previous findings suggesting that THP-1 monocytes exhibit low levels of MPO, whereas PMA-differentiated THP-1 macrophages display higher levels⁵⁰. Consequently, it is understandable that we did not observe CXCL1 upregulation following MPO treatment. However, it is noteworthy that MPO inhibitors, such as ABAH or Taurine, significantly decreased CXCL1 levels. Exploring alternative models, such as primary Kupffer cells or PMA-free macrophage cells, may offer an alternative approach to demonstrating MPO-induced CXCL1 expression.

CXCL1 is expressed in various liver cell types. Our studies revealed that CXCL1 treatment of HSCs or THP-1-derived macrophages significantly increased fibrogenesis and inflammatory markers, which were subsequently reduced by Atriol treatment. In co-culture studies, CXCL1 treatment of hepatocytes heightened THP-1-derived macrophage migration and inflammation marker expression, but these effects were nullified by Atriol treatment. Furthermore, factors secreted from CXCL1-treated Huh-7 cells induced fibrosis markers in LX-2, but Atriol treatment attenuated this induction. These findings underscore the significance of cell-to-cell interactions in simulating CXCL1's role in MASH pathogenesis, with Atriol mitigating these effects by reducing macrophage migration and fibrosis marker induction. Moreover, our in vitro cell lines and in vivo studies with primary hepatocytes isolated from GAN-induced MASH rats showed that Atriol protected mitochondrial function by inducing ATP production and maximal respiration. This finding was supported by data showing a reduction in NEFAs and induction of mitochondrial FAO genes of *PPARα* and *CPT1A* by Atriol in MASH. Importantly, Atriol increased FAO in MASH without compromising mitochondrial integrity, as evidenced by reduced ROS production.

MASH is complex even though FDA recently approved the first MASH drug³. Current drug development involves either a monotherapy targeting signal pathway or combination therapy targeting multiple pathways to enhance efficacy in MASH. Herein, we present compelling evidence demonstrating that Atriol is a promising single drug capable of targeting multiple pathways. While recognizing that Atriol might exert critical effects on specific cell types, potentially influencing others via paracrine mechanisms in vivo, we have refrained from pinpointing the primary target cell type. Nonetheless, if we are to assign Atriol's predominant effect, it would likely be its anti-inflammatory function. This is evidenced by the suppression of the downstream inflammatory target CXCL1, which not only compromised mitochondrial function in hepatocytes but also facilitated the transition from qHSC to aHSC.

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CRedit authorship contribution statement

Yuchang Li: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Liting Chen:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Chantal Sottas:** Writing – review & editing, Methodology, Investigation. **Mahima Chandrakant Raul, Nrupa Dinesh Patel:** Writing – review & editing, Methodology, Investigation, Data curation. **Janaki Ramulu Bijja:** Writing – review & editing, Methodology, Investigation, Data curation. **Syed Kaleem Ahmed:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Audrey Kapelanski-Lamoureux:** Writing – review & editing, Methodology. **Anthoula Lazaris, Peter Metrakos:** Writing – review & editing, Supervision, Methodology. **Alexander Zambidis:** Writing – review & editing, Investigation. **Shefali Chopra:** Methodology, Data curation. **Meng Li:** Methodology, Data curation. **Go Sugahara, Takeshi Saito:** Writing – review & editing, Resources. **Vassilios Papadopoulos:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

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Declaration of competing interest

Go Sugahara is an employee of PhoenixBio, Co., Ltd, Higashi-Hiroshima, Hiroshima, Japan.

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FIGURE LEGENDS

Fig.1. TSPO-ligand Atriol reverses the MASH phenotype in vitro.

(A) TSPO ligand 3, 17,19-androsten-5-trio (Atriol) chemical structure.

(B) Cellular thermal shift assay (CETSA) of Atriol binding TSPO in Huh-7 cells. Huh7 was treated with either DMSO (Veh) or Atriol (100µM). Illustration of the thermal aggregation curves with TSPO densitometry values normalized in immunoblots at various temperatures verse Veh at 37°C.

(C) Immunocytochemistry staining and quantification of nuclear NF-κB P65 in Huh-7 cells or PMA induced macrophages from THP-1 after treatment with DMSO (Veh) and different doses of Atriol (0µM, 10µM, 50µM, 100 µM). White arrows indicated nuclear NF-κB P65. P65 nuclear cell numbers were quantified using Image J. Scale, 20µm.

(D) Seahorse analysis of cellular oxygen consumption rate (OCR) of basal respiration, maximum respiration, and ATP production in humanized liver chimeric mouse-derived human hepatocytes (HLCM-HH) treated with Veh, LPS (100 mg/mL) and oleic acid (OA) (400 µM), or LPS+OA+Atriol (100 µM).

(E) Immunoblots of ACTA2, COL1A1, and GAPDH in LX-2 cells treated with Veh, TGFβ (1ng/ml), OCA (100nM), TGFβ+OCA, Atriol (100µM), or TGFβ+Atriol.

(F) Representative confocal images of COL1A1 (red) in LX-2 cells treated with Veh, TGFβ (1ng/ml), Atriol (100µM), or TGFβ+Atriol. Scale, 10µm. Data are expressed as mean ± SEM, *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001 by one-way ANOVA with Turkey's correction .

Fig. 2. Atriol ameliorates damage to liver structure in MASH.

(A) Schematic design of GAN diet-fed MASH rat model. During the final two weeks (week 41-43), Veh or Atriol (20mg/kg bw) was administered daily by intraperitoneal injection. PND, postnatal day.

(B) Representative images of the anatomy and livers stained with H&E, Oil Red O (ORO), or Sirius Red (SR) in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. Scale, 100µm.

(C) Liver weight/body weight ratio; ALT; AST; plasma cholesterol, insulin, non-esterified fatty acid (NEFA), and TAG; and hepatic TAG content in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups.

(D) H&E and F4/80 or Adiponectin immunofluorescent staining in white adipose tissue (WAT) in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. The white arrow indicates the thick membrane of the adipocytes; red arrowheads indicate crown-like adipocytes (CLA) in H&E staining slides. The red arrow indicates F4/80 positive crown-like structures, and the arrowhead indicates Adiponectin positive signals. Scale, 20µm. Data are expressed as mean ± SEM, *p < 0.05,

p < 0.01, *p < 0.001, ****p < 0.0001 by one-way ANOVA with Turkey's correction.

Fig. 3. Atriol reduces liver inflammation, fibrosis, cell death, de novo lipogenesis, and fatty acid transporter markers, while enhancing mitochondrial function indicators in MASH.

(A) Immunoblots were performed to assess the expression levels of CD11B, ACTA2, TGFβ, and cleaved caspase 3 (C-CASP3) in liver tissues in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. The fold change (FC) of CD11B, ACTA2, and TGFβ densities was calculated relative to the LFD+Veh group after normalization with GAPDH. The ratio of C-CASP3 to total caspase 3 (CASP3) was determined.

(B) Immunohistochemistry staining of F4/80, CD11B, NF- κ B, and ACTA2 in liver tissues in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. The white arrow indicates nuclear P65. Scale, 20 μ m.

(C) Confocal imaging and quantification of TUNEL+ cells after immunofluorescent staining of livers in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. White arrows indicate TUNEL positive cells. Scale, 10 μ m.

(D) qPCR of fibrosis markers (*Col1a1*, *Col1a2*, *Acta2*, *Tgf β* , and *Timp1*), inflammation markers (*Tnfa*, *Il1 β* , *Nlrp3*, *Cxcl1*, and *F4/80*), de novo lipogenesis (DNL) markers (*Acc1*, *Elov6*, *Fasn*, *Dgat1*, *Scd1*), fatty acid transporters (*Fabp5*, *Cd36*), and β -oxidation or mitochondrial biogenesis markers (*Ppara*, *Pgc1a*) in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. All experiments were repeated at least 3 times. Data are expressed as mean $\pm$ SEM, * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 by one-way ANOVA with Turkey's correction.

Fig. 4. Atriol maintains bile acid homeostasis and restores mitochondrial function without increasing ROS production in MASH in vivo.

(A) Immunoblot of FXR, PPAR α , CPT1A, and OXPHOS components in the livers in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups.

(B) Bile acid composition analysis in serum in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. GCA, glycocholic acid; GCDCA, glycochenodeoxycholic acid; GDCA, glycodeoxycholic acid; TDCA, taurodeoxycholic acid.

(C) Seahorse analysis of cellular oxygen consumption rate (OCR) of basal respiration, maximum respiration, and ATP production in Huh7 cells treated with Veh, GCA (200 μ M), GCDCA (200 μ M), GDCA (50 μ M), or TDCA (10 μ M) with or without Atriol.

(D) Seahorse analysis of basal respiration, maximum respiration, and ATP production in primary hepatocytes isolated from GAN+Veh, and GAN+Atriol rats.

(E) Immunoblot of CV-ATP5A, CIII-UQCRC2, FXR, PGC1 α , PPAR α , and TGF β in the primary hepatocytes from LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups.

(F) Representative images of immunofluorescence staining of 4-hydroxy-2-nonenal (HNE) and malondialdehyde (MDA) in liver in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. White arrow (top panels) indicates HNE signal and white arrowhead (bottom panels) indicates MDA signal. Scale, 20 μ m. All experiments were repeated at least 3 times. Data are expressed as mean $\pm$ SEM, * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 by one-way ANOVA with Turkey's correction in (B-C) and student's t-test (D).

Fig. 5. CXCL1 is downregulated by Atriol treatment in MASH.

(A) Plasma cytokine profiles and quantification in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups (n=3 pooled). (B) Immunoblot analysis of CXCL1, myeloperoxidase (MPO) from liver extracts in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups.

(C) Representative images of immunofluorescence staining of CXCL1 and MPO in liver in LFD+Veh, LFD+Atriol, GAN+Veh, and GAN+Atriol groups. The white arrow (top panels) and white arrowhead (bottom panels) indicate CXCL1 and MPO positive cells, respectively. Scale, 20 μ m.

(D) qPCR of CXCL1 in Huh-7, LX-2, and PMA induced macrophages from THP-1 cells after treatment with Veh, human recombinant MPO (1ng/ul), MPO+4-aminobenzoic acid hydrazide (4-ABAH) (200 μ M), or MPO+Taurine (50 μ M).

(E) Representative confocal images of CXCL1 (green) and COL1A1 (red) in LX-2 cells, or CXCL1 (red) and F4/80 (green) in PMA induced macrophages from THP-1 cells after treatment with Veh, MPO, MPO+ABAH or MPO+Taurine. Scale, 10 μ m.

(F) qPCR analysis of MPO expression in LX2 treated with human recombinant TGF β with or without Atriol.

(G) ELISA analysis of CXCL1 in the cultured medium from LX-2 cell (left) or in PMA induced macrophages from THP-1 cells (right) after treatment with Veh, recombinant MPO (1ng/ul), or MPO+ABAH (200mM).

(H) Colocalization analysis of CXCL1 (red) and Desmin (DES, green) (white arrows), or CXCL1 (red) and F4/80 (green) (white arrowhead) in rat livers in GAN+Veh and GAN+ABAH. Scale, 20 μ m.

(I) Immunoblot analysis and immunofluorescent staining of ACTA2. White arrow indicates ACTA2 signal (green). Scale, 20 μ m. Data are expressed as mean $\pm$ SEM, * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 by one-way ANOVA with Turkey's correction in (B, D, F) and student's t-test (H).

Fig. 6. CXCL1 impairs mitochondrial function, activates HSCs, induces inflammation, and promotes macrophage infiltration, whereas Atriol counters these effects.

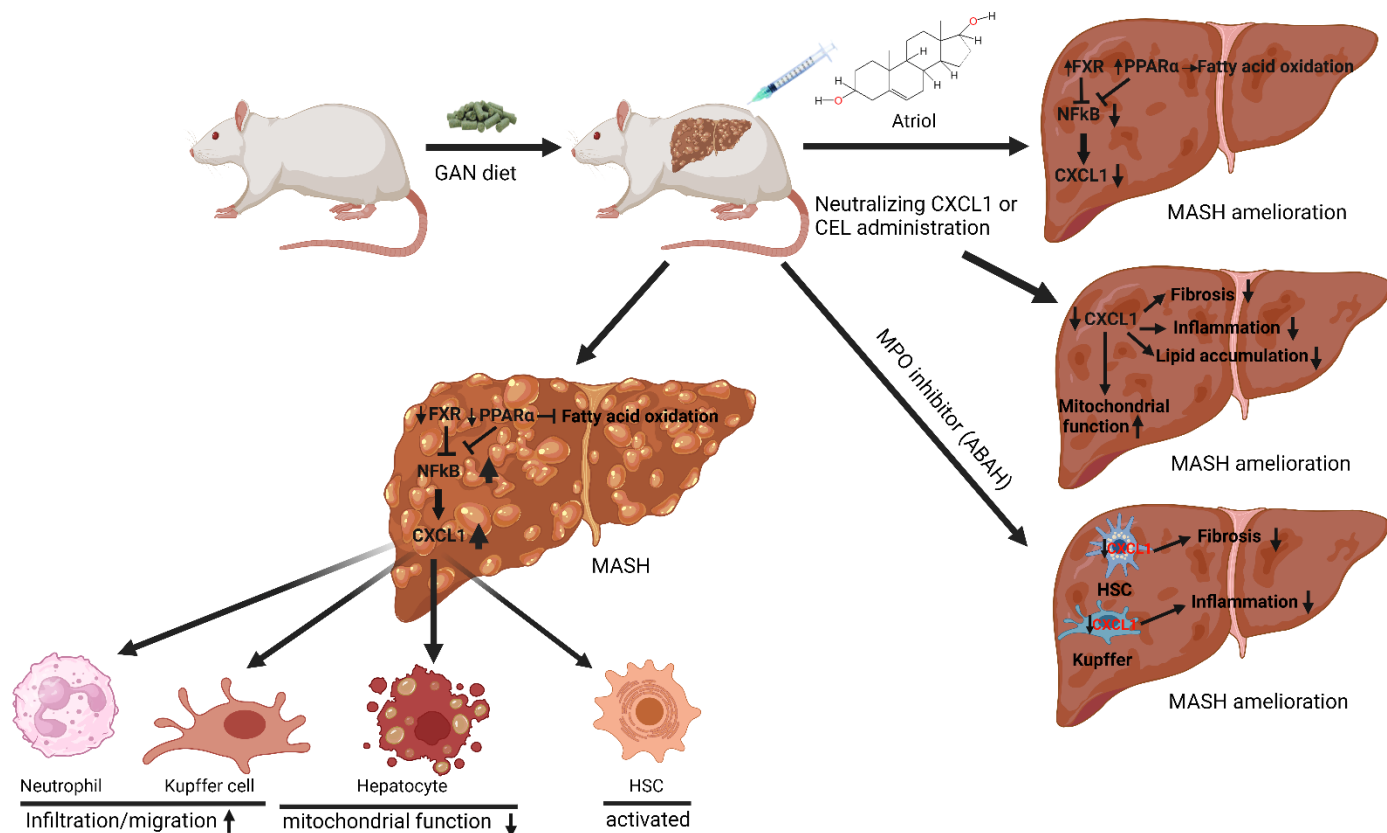
(A) H&E staining of livers from normal, SS, or MASH patients, black arrow indicates infiltration inflammation. Scale, 100 μ m. CXCL1 colocalization with ASGR1, CD68, or DESMIN (DES) in liver; white arrow indicates colocalization. Scale, 100 μ m.

(B) Seahorse analysis in HLCM-HH treated with Veh, CXCL1 (100 ng/mL), or CXCL1+ Atriol (100uM).

- (C) Immunoblot analysis of ACTA2 and COL1A1 in human LX-2 cells or rat BSC treated with Veh, CXCL1, or CXCL1+Atritol.
- (D) Representative confocal images of immunofluorescence staining of COL1A1 in LX-2 cells or BSC treated with Veh, CXCL1, or CXCL1+ Atritol. Scale, 10 μ m.
- (E) Immunoblot and quantification of TNF α in THP-1-derived macrophages treated with Veh, CXCL1, or CXCL1+Atritol.
- (F) Representative confocal images after immunofluorescence staining of TNF α in THP-1-derived macrophages treated with Veh, CXCL1, or CXCL1+ Atritol; the red color indicates TNF α positive signals. Scale, 10 μ m.
- (G) Co-culture of Huh-7 cells in lower chamber treated with Veh, CXCL1, or CXCL1+ Atritol, and THP-1-derived macrophages in upper chamber. Migrated cells in the upper chamber were stained and quantified. Scale, 100 μ m. Data are expressed as mean $\pm$ SEM, * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 by one-way ANOVA with Turkey's correction in.

Fig. 7. Pharmacological inhibition of NF- κ B/CXCL1 axis or neutralizing CXCL1 effectively reproduces the mitigating effects of Atritol on MASH.

- (A) Immunoblot analysis of TNF α , CXCL1, and ACTA2 from liver extracts in GAN+Veh and GAN+CEL (Celastrol).
- (B) Immunohistochemistry staining of NF- κ B (black arrow indicates nuclear NF- κ B), F4/80 (white arrow), and immunofluorescence staining of CD11B (white arrow), ACTA2 (white arrow) in the livers in GAN+Veh and GAN+CEL. Scale, 20 μ m.
- (C) Immunoblot analysis of NF- κ B p65 from nuclear and cytoplasmic fractions from primary hepatocytes isolated from GAN+Veh and GAN+CEL rats (n=3).
- (D) Immunoblot of TNF α , IL1 β , ACTA2, FXR, and PPAR α in rat liver extracts in LFD, GAN+Veh, and GAN+CXCL1 neu.
- (E) Immunohistochemistry staining of ACTA2, F4/80 (white arrow), or NF- κ B p65 (red arrowhead indicates nuclear localization) in livers in LFD, GAN+Veh, and GAN+CXCL1 neu. Scale, 20 μ m.
- (F) TBARS levels in rat liver tissues from LFD, GAN+Veh, and GAN+CXCL1 neu groups. Data are expressed as mean $\pm$ SEM, * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 by one-way ANOVA with Turkey's correction in (F) and student's t-test (A, C).



Graphical abstract

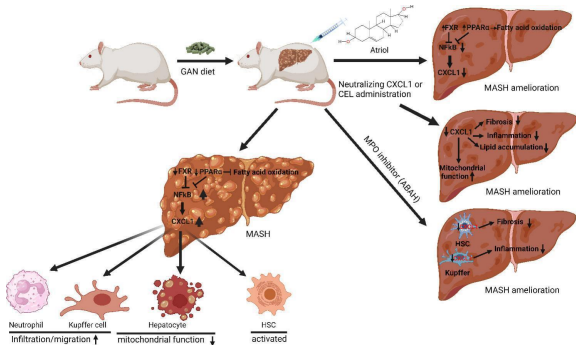
Atriol, a TSPO ligand, restores the dysfunctional mitochondria, reverses the activated HSCs, and prevents inflammation by blocking the NF- κ B pathway in vitro.

CXCL1 impairs the mitochondrial function, activates hepatic stellate cells (HSCs), induces inflammation, and promotes macrophage migration.

CXCL1 is upregulated in MASH in patients and rats, while the neutralization of CXCL1 ameliorates MASH in rats.

MPO regulates CXCL1 in HSCs and Kupffer cells but not in hepatocytes, while Atriol downregulates MPO.

Atriol downregulates CXCL1 to ameliorate MASH.



Graphics Abstract

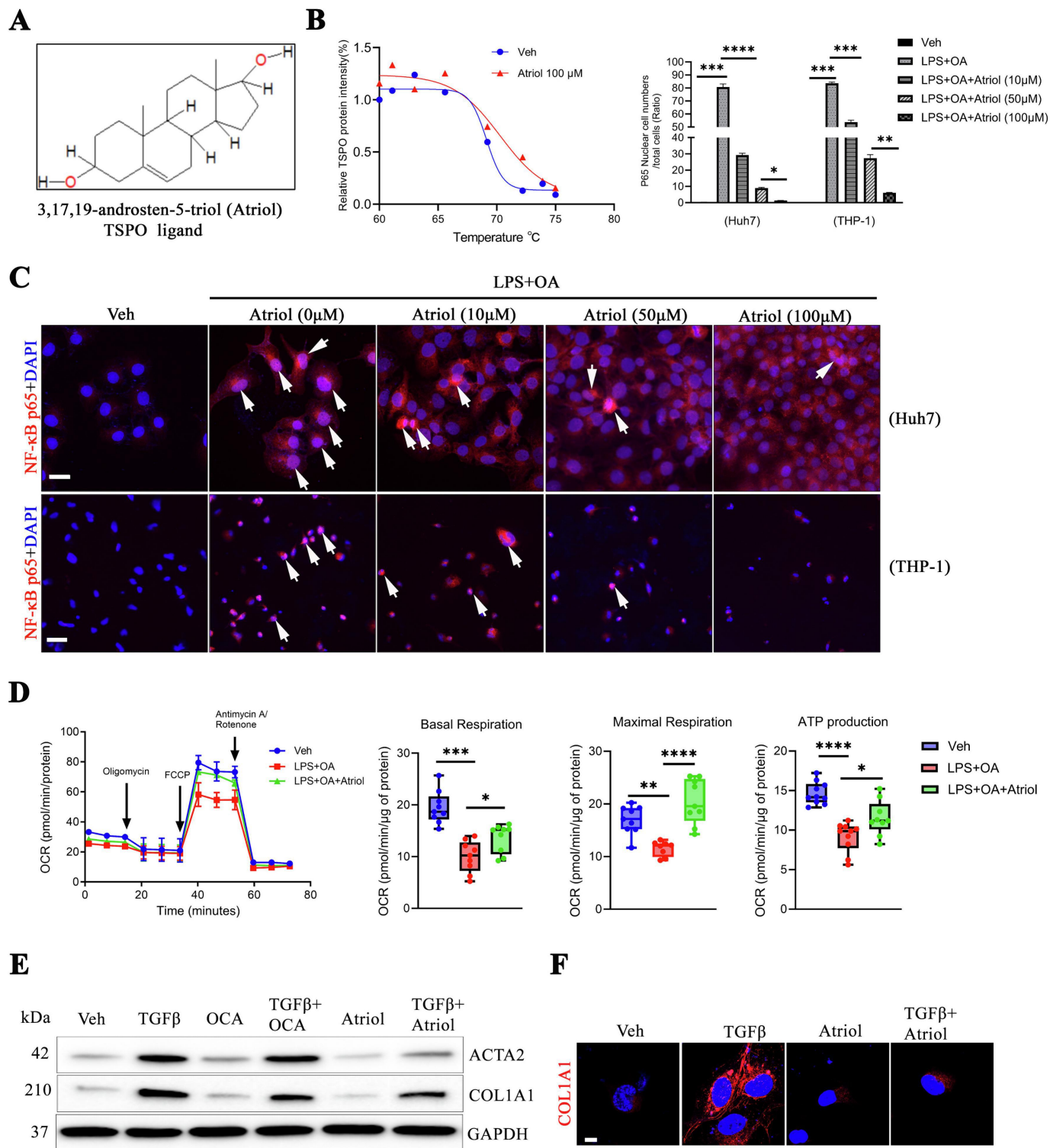


Figure 1

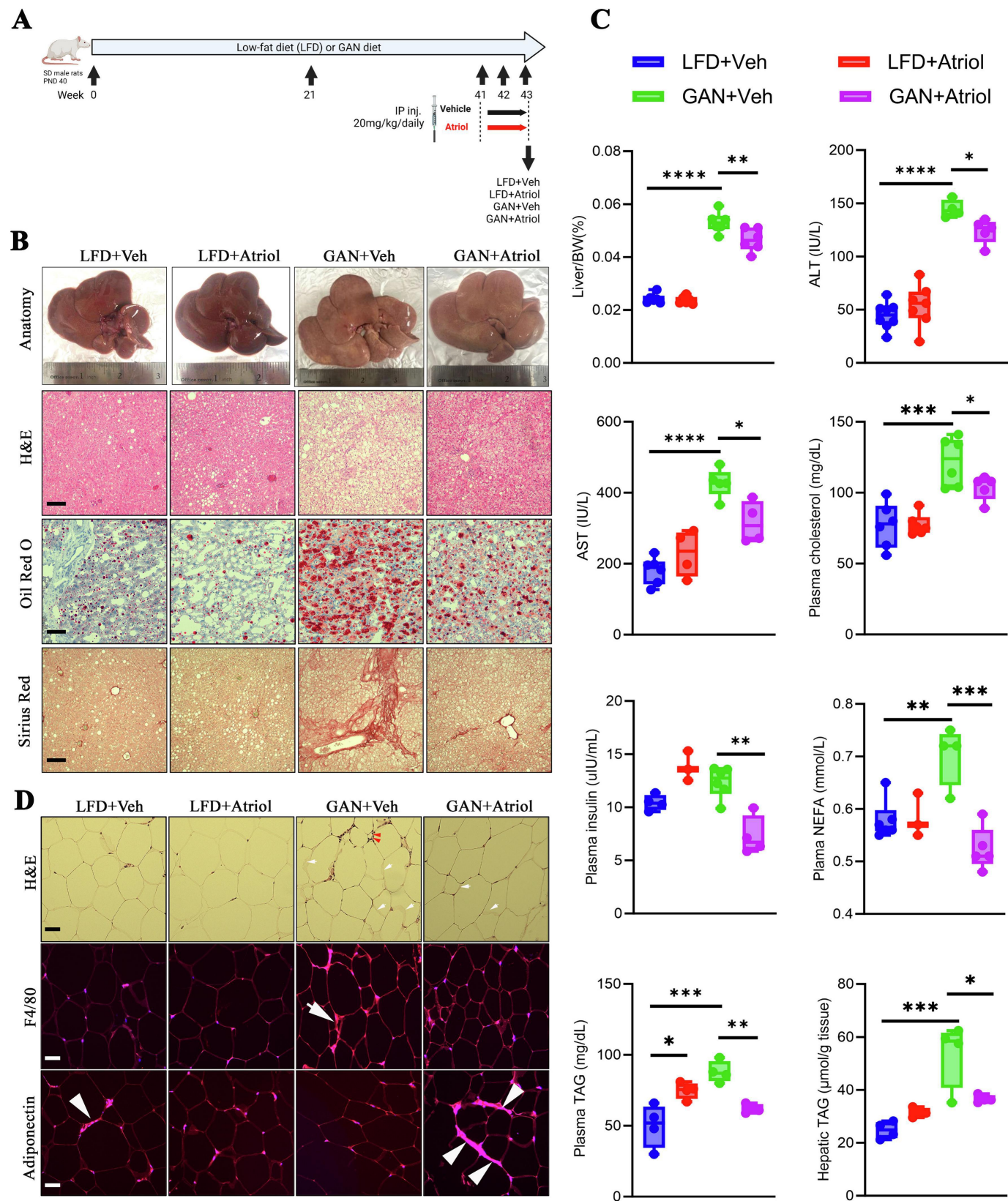


Figure 2

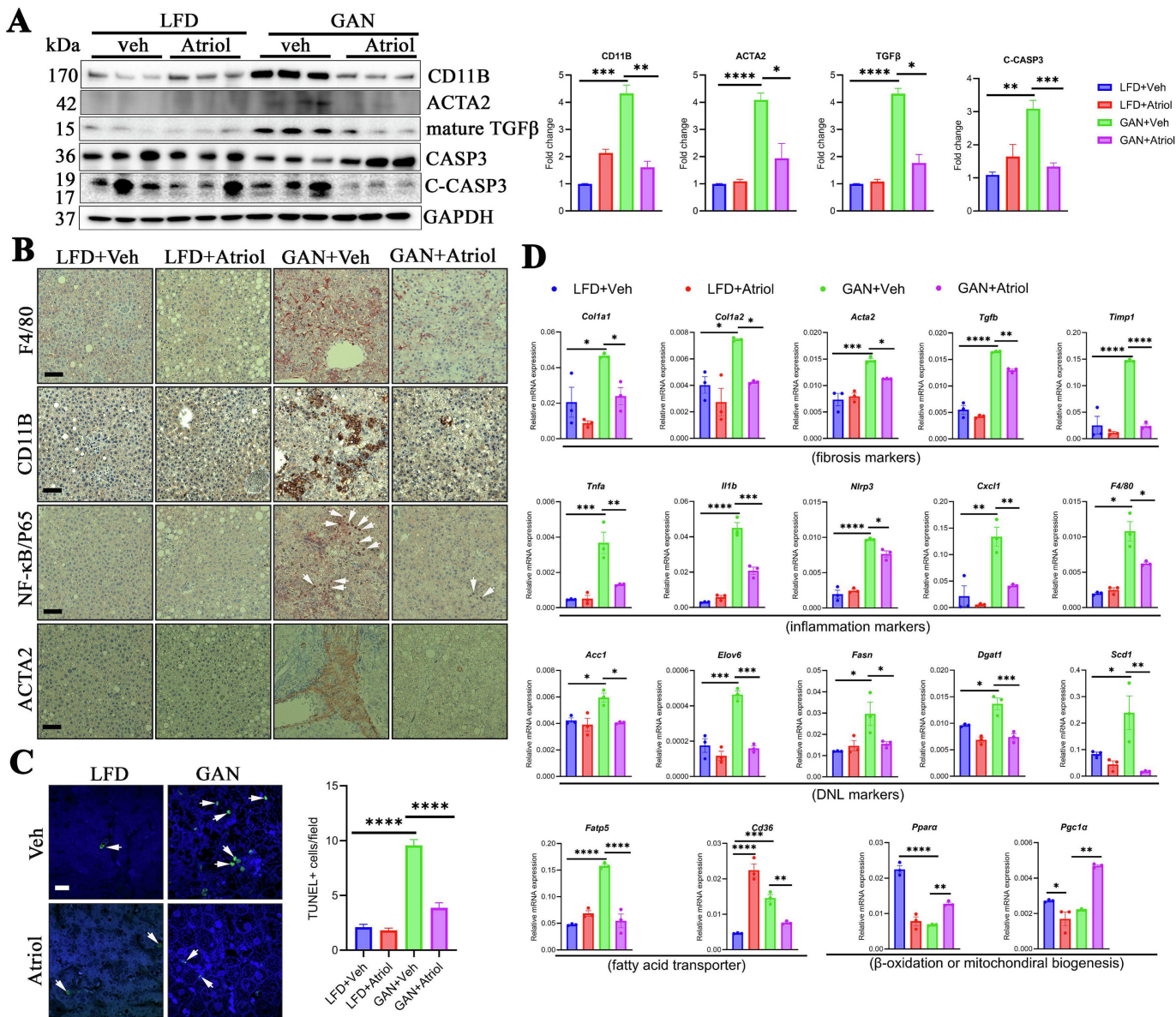


Figure 3

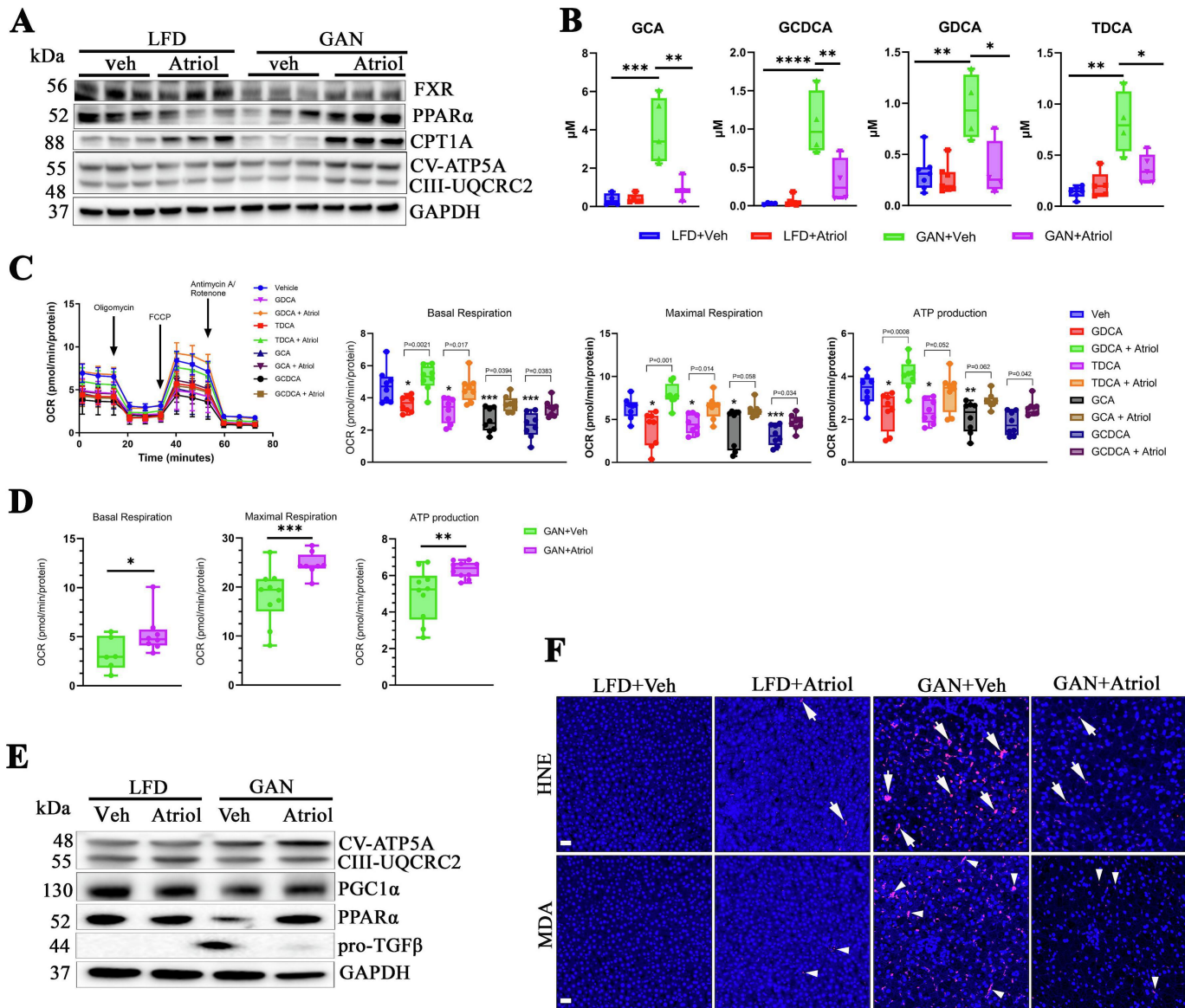


Figure 4

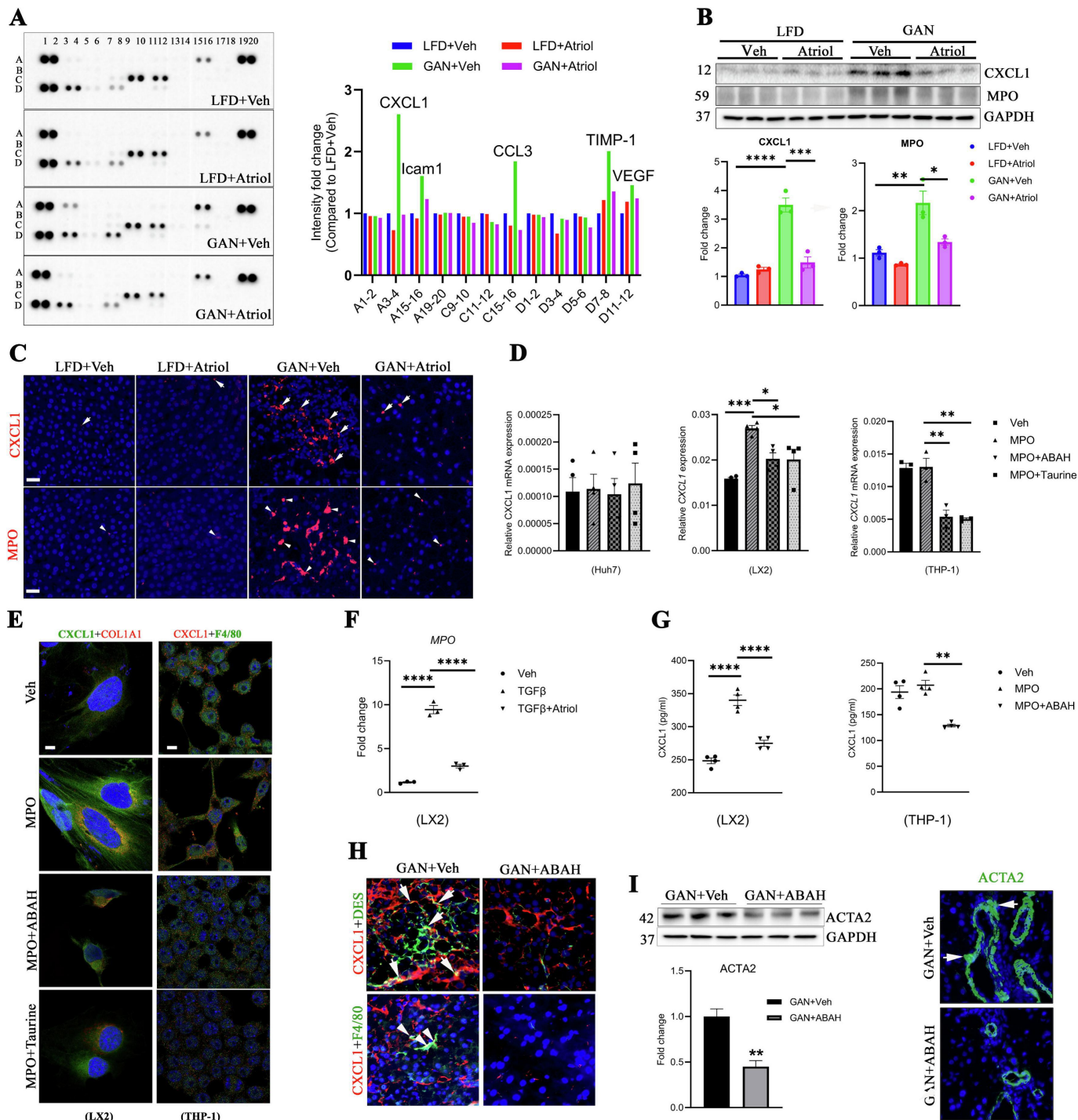


Figure 5

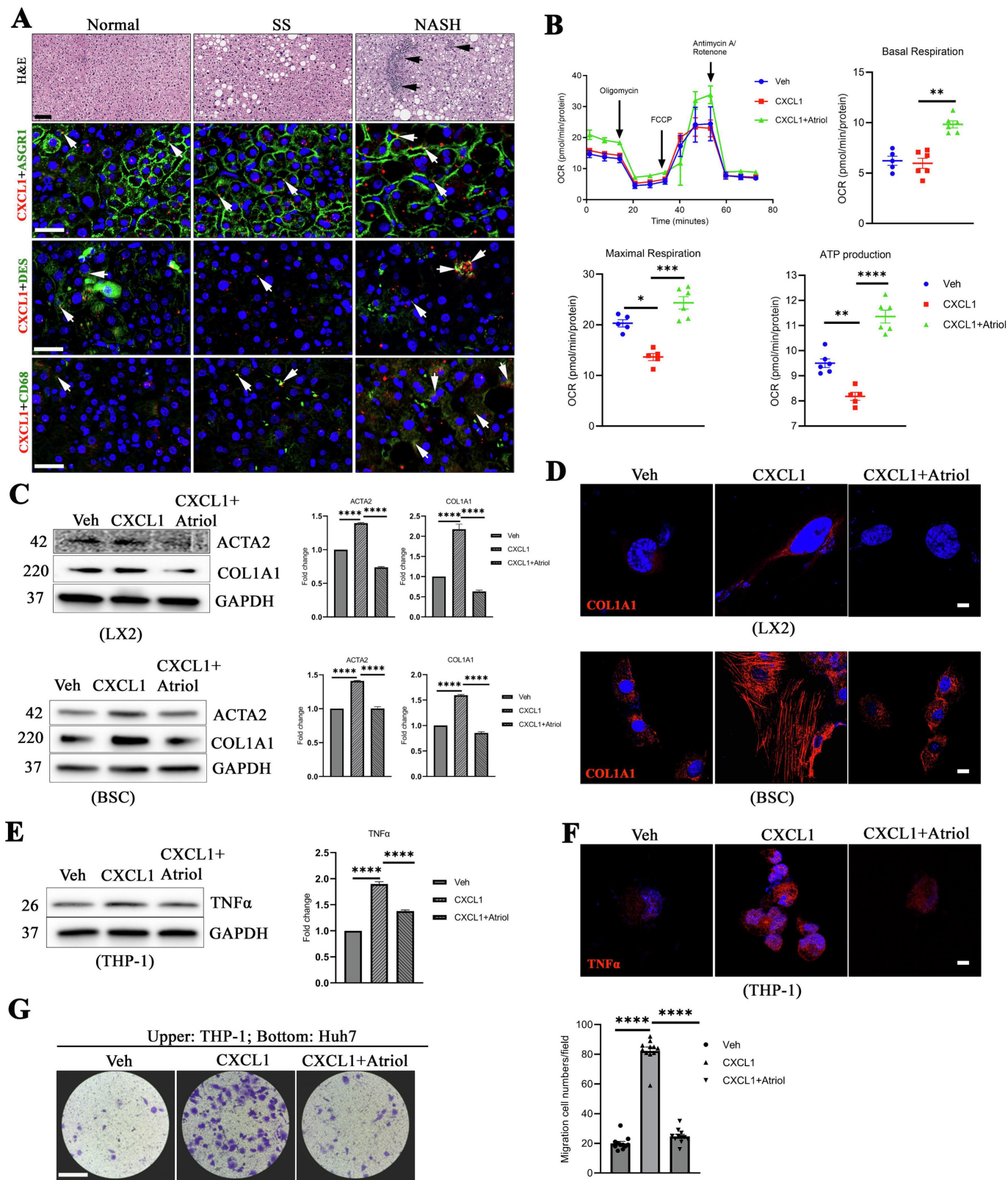


Figure 6

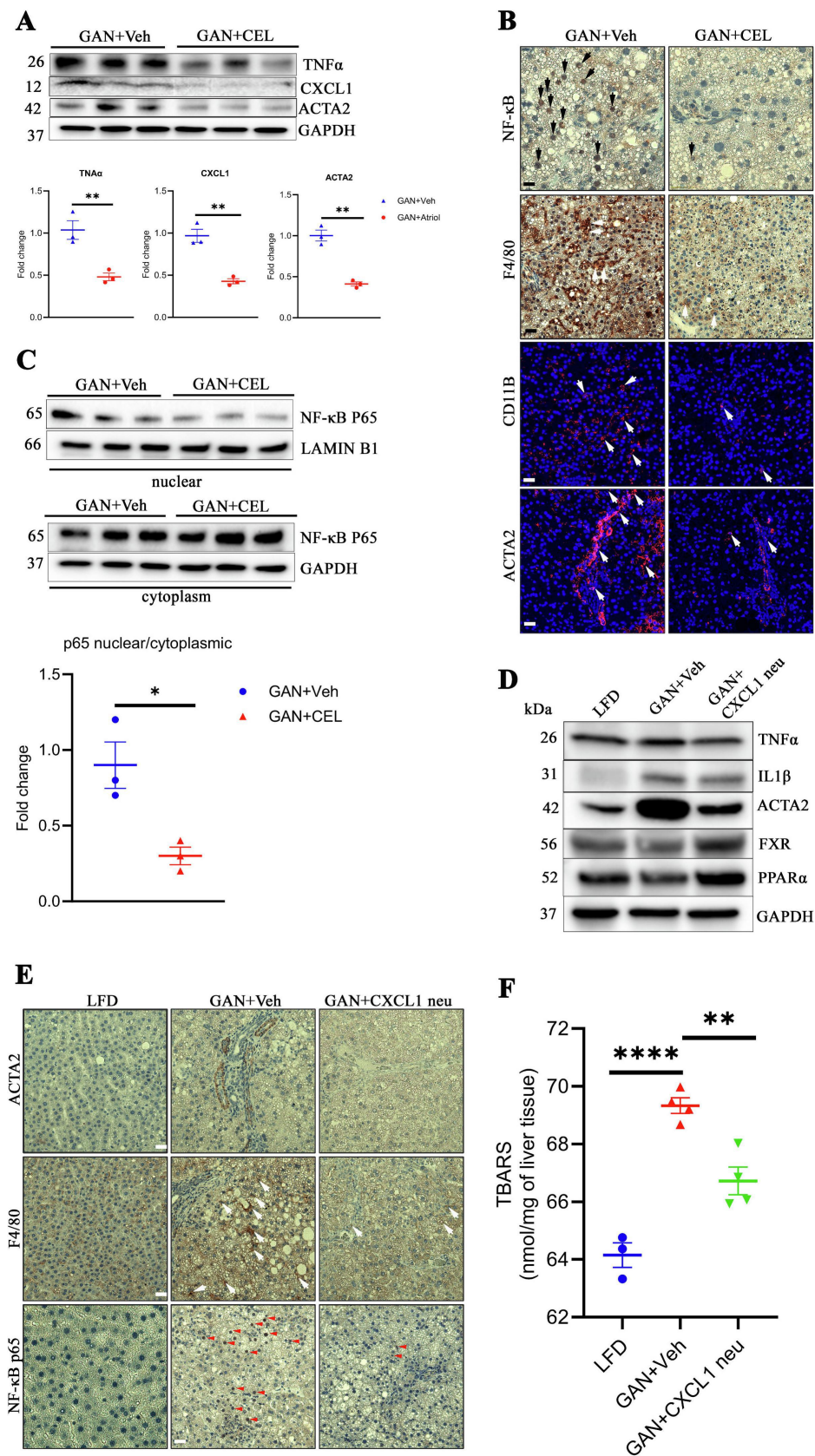


Figure 7